



**CERTUS**  
CONTINUOUS MEDIA PROVENANCE

# **CERTUS – CONTINUOUS MEDIA PROVENANCE PRODUCT REQUIREMENTS DOCUMENT (PRD)**

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# INDEX

## Table of Contents

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| <b>1. Product Foundation .....</b>                              | <b>5</b>  |
| 1.1 Executive Summary .....                                     | 5         |
| 1.2 Background .....                                            | 5         |
| 1.3 Product Ideation .....                                      | 5         |
| 1.4 Problem Statement.....                                      | 7         |
| 1.5 Product Vision .....                                        | 8         |
| 1.6 Product Objectives .....                                    | 8         |
| 1.7 Product Principles .....                                    | 8         |
| 1.8 Customer Value Proposition .....                            | 8         |
| 1.9 Value Proposition Canvas .....                              | 9         |
| 1.10 Compelling Factor for the Customer .....                   | 10        |
| <b>2. Product Design .....</b>                                  | <b>10</b> |
| 2.1 Branding.....                                               | 10        |
| 2.2 Design Language System.....                                 | 11        |
| 2.3 Jobs To Be Done (JTBD) .....                                | 11        |
| 2.4 User Personas .....                                         | 12        |
| 2.5 User Stories.....                                           | 12        |
| 2.6 User Journeys .....                                         | 13        |
| 2.7 User Flows .....                                            | 14        |
| 2.8 Accessibility and Responsive Design .....                   | 14        |
| <b>3. Technical Architecture.....</b>                           | <b>14</b> |
| 3.1 Cryptography Glossary and Justifications .....              | 14        |
| 3.2 Technical Architecture Overview .....                       | 15        |
| 3.3 Architectural Design Principles .....                       | 15        |
| 3.4 End to End Workflow .....                                   | 15        |
| 3.5 Source Media and Telephony.....                             | 15        |
| 3.6 Provenance Generation and Cryptographic Fingerprinting..... | 15        |
| 3.6.1 Chunk Construction .....                                  | 15        |
| 3.6.2 Final Partial Chunk Handling .....                        | 15        |
| 3.6.3 Provenance Binding.....                                   | 15        |

|                                                                              |           |
|------------------------------------------------------------------------------|-----------|
| 3.6.4 SHA-256 Chunk Fingerprinting .....                                     | 15        |
| 3.6.5 Ordered Chunk Fingerprints .....                                       | 15        |
| 3.6.6 Provenance Stage Output .....                                          | 15        |
| <b>3.7 Source Merkle Root.....</b>                                           | <b>16</b> |
| 3.7.1 Merkle Tree Input.....                                                 | 16        |
| 3.7.2 Merkle Tree Construction .....                                         | 16        |
| 3.7.3 Odd Number of Hashes .....                                             | 16        |
| 3.7.4 Why Certus Generates a Source Merkle Root .....                        | 16        |
| 3.7.5 Why the Source Merkle Root Is Carried Forward .....                    | 16        |
| 3.7.6 What the Source Merkle Root Does Not Establish .....                   | 16        |
| 3.7.7 Merkle Stage Output .....                                              | 16        |
| <b>3.8 Transformations and Signed Attestations .....</b>                     | <b>16</b> |
| 3.8.1 Transformation Chain .....                                             | 16        |
| 3.8.2 Transformation Attestation .....                                       | 17        |
| 3.8.3 Transformation Continuity .....                                        | 17        |
| 3.8.4 Digital Signing and Signature Verification .....                       | 17        |
| 3.8.5 Scope of a Valid Digital Signature .....                               | 17        |
| 3.8.6 Final Expected Media State and Expected Final Hash .....               | 17        |
| <b>3.9 Independent Verification and Classifications.....</b>                 | <b>17</b> |
| 3.9.1 Independent Verification.....                                          | 17        |
| 3.9.2 Root-Cause Analysis .....                                              | 17        |
| 3.9.3 Integrity Classification .....                                         | 17        |
| 3.9.4 Verification Record .....                                              | 17        |
| <b>3.10 Orchestration, APIs and Persistence.....</b>                         | <b>17</b> |
| 3.10.1 n8n Orchestration.....                                                | 18        |
| 3.10.2 Backend API Layer .....                                               | 18        |
| 3.10.3 Workflow Data Continuity .....                                        | 18        |
| 3.10.4 Persistence .....                                                     | 18        |
| <b>3.11 Front End Integration.....</b>                                       | <b>18</b> |
| <b>3.12 Testing Plan .....</b>                                               | <b>18</b> |
| <b>3.13 Acceptance Criteria .....</b>                                        | <b>18</b> |
| <b>3.14 Known Limitations .....</b>                                          | <b>18</b> |
| <b>3.15 Enterprise Readiness: Reliability, Security and Compliance .....</b> | <b>18</b> |
| <b>4. Product Strategy.....</b>                                              | <b>18</b> |
| 4.1 Segmentation, Targeting and Positioning (STP).....                       | 18        |
| 4.2 Lean Model Canvas .....                                                  | 19        |
| 4.3 Product Metrics .....                                                    | 20        |
| 4.4 Feature Prioritisation .....                                             | 21        |
| 4.5 Build, Buy, Integrate Strategy .....                                     | 22        |
| 4.6 Customer Service Strategy .....                                          | 22        |
| <b>5. Marketing Plan .....</b>                                               | <b>23</b> |
| <b>6. Ethical Considerations.....</b>                                        | <b>23</b> |

**7. Future Roadmap.....23**

**8. Project Management.....24**

## 1. Product Foundation

### 1.1 Executive Summary

Certus is an observational provenance layer that continuously records and verifies speech integrity across AI-processed speech pipelines. It establishes a verifiable record of speech transformations, validates transformation continuity and classifies integrity outcomes across the media lifecycle. The MVP explores how continuous provenance can provide an independent accountability layer across increasingly complex AI speech pipelines.

### 1.2 Background

Contact-centre speech is increasingly processed by AI systems for noise cancellation, source separation, accent conversion and other enhancements. These legitimate transformations intentionally alter the audio before it reaches listeners and downstream systems. As a result, a change in speech can no longer by itself distinguish a legitimate transformation from channel effects or an unexplained alteration. Certus was conceived to explore this emerging provenance gap.

### 1.3 Product Ideation

At the macro level, the **PESTLE framework** was used to assess the environment around AI-transformed speech. At the micro level, the **ERRC framework** was used to define the product opportunity. This led to the product hypothesis that continuous, transformation-aware provenance could provide an independent accountability layer across AI speech pipelines.

PESTLE Framework is as follows:

| S.No | PESTLE Framework | Explanation                                                                                                                                                                                 |
|------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Political        | As of 05 September 2026, no specific political barrier has been identified in India that prevents the development or deployment of a media-provenance product such as Certus.               |
| 2    | Economic         | Increased investment in AI-enabled voice infrastructure creates a growing need to manage the operational and audit costs associated with increasingly complex speech-processing ecosystems. |
| 3    | Social           | Increasing use of AI to modify human speech raises broader concerns around authenticity, transparency and trust in voice interactions.                                                      |
| 4    | Technological    | Advances in real-time speech transformation, cryptography and cloud infrastructure make continuous, transformation-aware media provenance technically feasible.                             |
| 5    | Legal            | Certus deployments must comply with applicable provisions of India's DPDP framework, CERT-In requirements, and other                                                                        |

## CERTUS PRD

|   |               |                                                                                                                                                                                      |
|---|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |               | relevant cybersecurity, telecommunications, privacy and sector-specific laws and regulations.                                                                                        |
| 6 | Environmental | Certus should optimise its environmental footprint through compute-efficient verification, efficient media retention and responsible use of storage infrastructure as Certus scales. |

ERRC Framework is as follows:

| S.No | ERRC             | Summary                                                                                                    |
|------|------------------|------------------------------------------------------------------------------------------------------------|
| 1    | <b>Eliminate</b> | Binary assumption that any changed waveform is automatically malicious.                                    |
| 2    | <b>Reduce</b>    | Ambiguity in tracing integrity across multiple speech-processing stages and providers.                     |
| 3    | <b>Raise</b>     | Auditability, transformation-provider accountability and explainability of integrity outcomes.             |
| 4    | <b>Create</b>    | A continuous provenance layer linking source speech, transformations, attestations and final verification. |

## SCAMPER Framework

Within the SCAMPER framework, Certus aligns with both **Combine** and **Adapt**.

**Combine** - Certus combines established cryptographic and provenance mechanisms, including SHA-256 fingerprinting, Merkle commitments and digital signatures, with AI speech-processing workflows to establish continuous provenance across successive transformations.

**Adapt** - Certus adapts these established cryptographic and provenance mechanisms to the context of AI-processed contact-centre speech, where media undergoes multiple transformations.

**Assumption:** For this analysis, **Speech Enhancement** encompasses noise suppression/cancellation, acoustic echo cancellation, background-voice suppression, dereverberation, beamforming and related audio-restoration techniques. **Speech Transformation** refers to intentional modification of speech characteristics, including accent conversion and voice conversion.

| S.No | Area                               | Conclusion                                                                                                                                                                                   | Source                                                   |
|------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1    | <b>Speech Processing &amp; ASR</b> | Noise removal can suppress phonetic/acoustic cues required by ASR. ASR-optimised processing demonstrates that upstream transformations must preserve downstream-relevant speech information. | <a href="#">Sanas — ASR-Optimized Noise Cancellation</a> |

|   |                                                    |                                                                                                                                                                                                           |                                                                                                          |
|---|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 2 | <b>Speech Processing &amp; Speaker Recognition</b> | Speech enhancement can attenuate genuine speech and partially lose speaker information. Modern restoration techniques seek to mitigate this loss, demonstrating continued upstream/downstream dependency. | <a href="#">IEEE — Attenuated Speech Restoration for Speaker Verification</a>                            |
| 3 | <b>Accent Conversion &amp; Speaker Identity</b>    | Accent conversion can alter speaker identity alongside accent. Modern systems therefore use explicit speaker-preservation mechanisms to mitigate identity alteration.                                     | <a href="#">Sanas SDK — November 2025 · IEEE — Voice-Preserving Zero-Shot Multiple Accent Conversion</a> |
| 4 | <b>Caller Authentication</b>                       | CallShield demonstrates real-time audio-layer caller authentication via neural watermarking, but does not maintain provenance across AI transformation stages.                                            | <a href="#">CallShield</a>                                                                               |
| 5 | <b>Tamper Verification</b>                         | SpeechVerifier distinguishes malicious modification from benign processing via fingerprinting and segment-wise watermarking, but does not maintain continuity across AI transformations.                  | <a href="#">SpeechVerifier</a>                                                                           |
| 6 | <b>Speech Provenance</b>                           | MerkleSpeech demonstrates cryptographically verifiable, chunk-level speech provenance, but verifies against a signed source rather than a multi-transformation chain.                                     | <a href="#">MerkleSpeech</a>                                                                             |
| 7 | <b>Media Provenance</b>                            | C2PA demonstrates that media history and transformations can be represented through cryptographically verifiable provenance.                                                                              | <a href="#">C2PA Specification</a>                                                                       |

As AI speech pipelines become increasingly layered, transformations can alter information relied upon by downstream systems such as ASR and speaker verification. This presents an emerging product opportunity: Continuous Media Provenance across AI-transformed speech pipelines. Existing approaches reviewed address caller authentication, tamper verification and source-level speech provenance, indicating an opportunity to explore continuous provenance across AI speech transformations and subsequent media changes.

#### 1.4 Problem Statement

Based on the product and technical research, the problem statement identified is:

**Continuous, verifiable provenance is an emerging challenge for AI-transformed contact-centre speech, as legitimate transformations and ordinary telephony effects can both alter the media, thus obscuring its transformation history and integrity.**

### 1.5 Product Vision

To establish continuous media provenance as a trust layer for AI-transformed enterprise speech, enabling every transformation to remain verifiable, attributable and auditable throughout the speech lifecycle.

### 1.6 Product Objectives

**The primary objective of Certus is to demonstrate the technical feasibility of continuous media provenance across AI speech transformations.** The MVP establishes source provenance, records and verifies transformations through signed attestations, and evaluates final received media using cryptographic and contextual channel evidence.

The long-term objective is to establish Certus as a production-grade, real-time provenance layer for enterprise speech, with provider integrations, scalable verification, policy enforcement and evidentiary reporting.

### 1.7 Product Principles

The product principles are:

- i. **Evidence before inference** - Cryptographic and transformation evidence should be evaluated before assigning an integrity classification.
- ii. **Authorised change is not manipulation** - Legitimate AI transformations must remain distinguishable from unauthorised alteration.
- iii. **Continuous provenance** - Provenance should be preserved across the speech lifecycle, rather than verified only at the final output.
- iv. **Provider accountability** - Every transformation should be attributable to a provider, method and signed attestation.
- v. **Role-based transparency** - Contact centres receive relevant verification evidence, while Certus Administrators have access to detailed provenance and integrity information.
- vi. **Modular verification** - Validation, provenance, transformation, attestation, verification and root-cause analysis remain independently testable.

### 1.8 Customer Value Proposition

The value proposition statement is:

**For [contact centres and AI speech-processing providers],  
who want to [use AI-processed speech while preserving verifiability and accountability],**

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Certus is an [observational provenance layer for AI-processed speech pipelines], that helps them [verify transformations and investigate unexplained media changes], by [linking source provenance, transformation-provider attestations and downstream verification into a continuous, auditable chain].

The elevator pitch statement is:

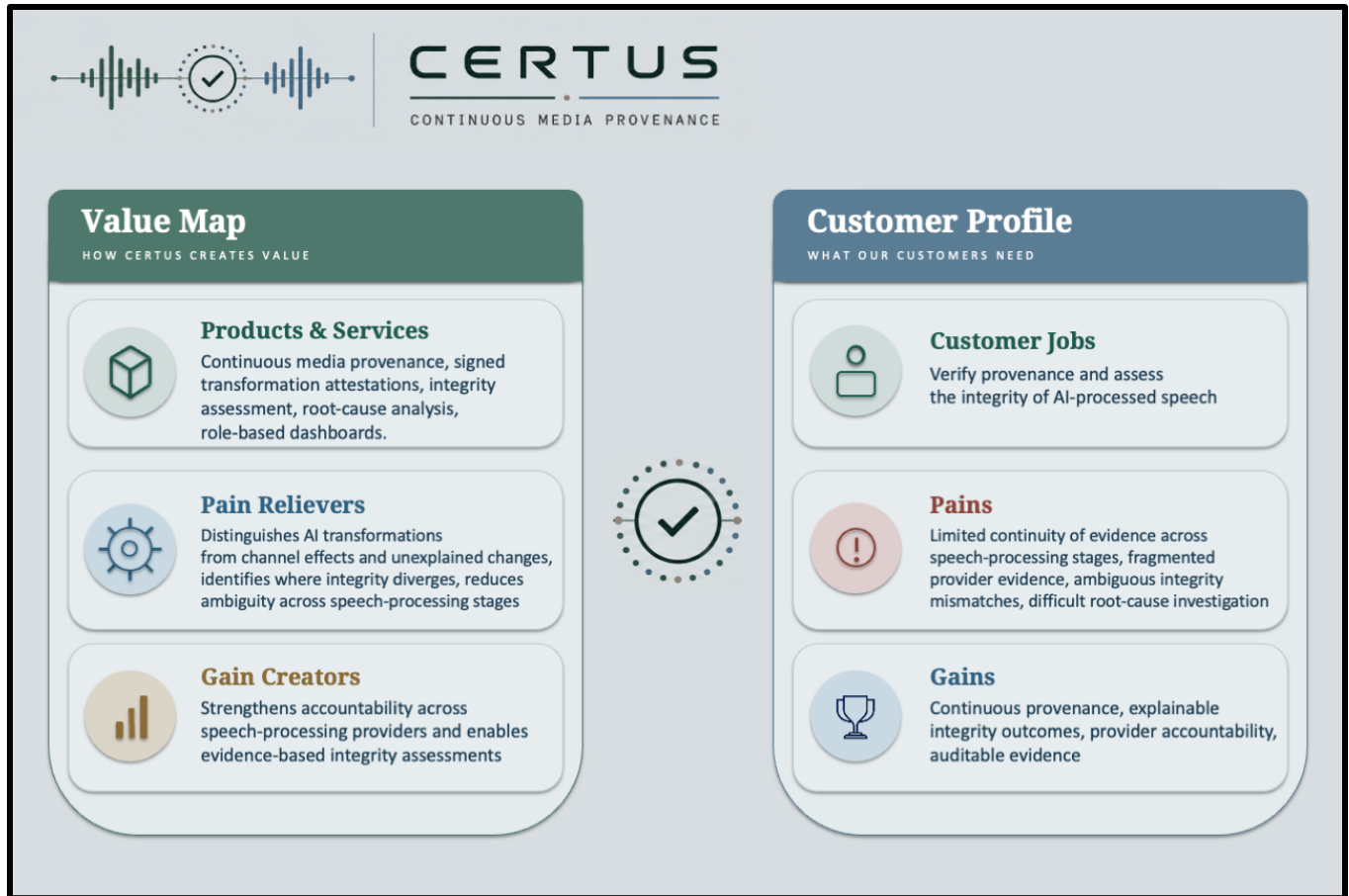
**Certus provides continuous media provenance and integrity verification across the lifecycle of AI-processed speech.**

### 1.9 Value Proposition Canvas

The value proposition table is as follows:

| Section          | Category            | Details                                                                                                                                                               |
|------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer Profile | Customer Jobs       | Verify provenance and assess the integrity of AI-processed speech                                                                                                     |
|                  | Pains               | Limited continuity of evidence across speech-processing stages, fragmented provider evidence, ambiguous integrity mismatches, difficult root-cause investigation      |
|                  | Gains               | Continuous provenance, explainable integrity outcomes, provider accountability, auditable evidence                                                                    |
| Value Map        | Products & Services | Continuous media provenance, signed transformation attestations, integrity assessment, root-cause analysis, role-based dashboards.                                    |
|                  | Pain Relievers      | Distinguishes AI transformations from channel effects and unexplained changes, identifies where integrity diverges, reduces ambiguity across speech-processing stages |
|                  | Gain Creators       | Strengthens accountability across speech-processing providers and enables evidence-based integrity assessments                                                        |

The value proposition canvas image is as follows:



**Fig 1.9.1: Customer Value Proposition Canvas**

## 1.10 Compelling Factor for the Customer

As AI speech passes through multiple processing stages and providers, maintaining a reliable record of its provenance becomes an emerging need. Certus addresses this through continuous media provenance, enabling clearer integrity assessment and accountability. The business value remains to be validated through customer research.

## 2. Product Design

### 2.1 Branding

Certus is derived from the Latin word associated with certainty, assurance and trustworthiness. The name reflects the product's purpose: establishing confidence in media that has legitimately changed. The logo combines a speech waveform with a verification mark to represent transformed audio and provable trust. The restrained grey-green-blue visual identity is intended to communicate security, technical confidence and enterprise reliability without resembling a consumer audio product.



**Fig 2.1.1: Light Themed Logo**

The logo is designed to remain legible across light enterprise interfaces and supporting documentation. Alternate dark-theme applications may be developed later while preserving the core waveform and verification motif.

## 2.2 Design Language System

Certus follows a restrained enterprise design language centred on trust, verification and evidence. The interface uses pale grey-blue backgrounds (#E7ECEC, #F3F6F6), deep green primary typography (#1B3B31) and muted status colours to communicate integrity outcomes: green (#55776A) for Authorised Transformation, blue (#6F8FAF) for Benign Channel Variation, amber (#A7865D) for Integrity Indeterminate and muted red (#A45454) for Suspected Unauthorised Manipulation. Cormorant Garamond is used for primary headings and Inter for operational content, with monospaced typography reserved for hashes and technical identifiers. Clean spacing, subtle borders and restrained visual hierarchy keep evidence-dense screens readable, while role-based views provide relevant verification evidence to Contact Centre users and deeper provenance and integrity information to Certus Administrators.

## 2.3 Jobs To Be Done (JTBD)

The 2 main JTBD have been identified as follows:

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### 1. JTBD - Maintain Continuous Speech Provenance

**When** speech passes through multiple AI-processing stages and providers,  
**I want to** maintain a verifiable provenance record across the speech lifecycle,  
**So that** its processing history remains traceable and auditable.

**Emotional Drivers:** Confidence in the provenance record; reduced uncertainty across complex speech pipelines.

### 2. JTBD - Assess Media Integrity

**When** the received media differs from the expected outcome,  
**I want to** assess the available provenance, transformation and channel evidence,  
**So that** I can understand the integrity outcome without treating every media change as manipulation.

**Emotional Drivers:** Clarity during integrity exceptions; confidence in evidence-based assessments.

## 2.4 User Personas

The user personas are:

| Persona                      | Goals                                             | Pain Points                                                    | Success Metric                                    |
|------------------------------|---------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------|
| Contact Centre Administrator | Verify provenance and assess call integrity       | Limited evidence continuity, ambiguous integrity mismatches    | Clear, explainable integrity outcomes             |
| Certus Administrator         | Monitor provenance and integrity across customers | Complex evidence across calls, providers and processing stages | Accurate, evidence-based integrity classification |

## 2.5 User Stories

The main user stories are:

- i. Persona 1 - Contact Centre Administrator
  - a. User Story 1 - As a Contact Centre Administrator, I want to view the provenance and integrity outcomes for my organisation's calls, so that I can understand the evidence behind each integrity assessment.
  - b. User Story 2 - As a Contact Centre Administrator, I want to view the transformation providers and methods associated with each call, so that I can trace how the media was processed across its lifecycle.

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- ii. Persona 2 - Certus Administrator
  - a. User Story 1 - As a Certus Administrator, I want to review provenance and integrity records across customers, so that I can investigate integrity exceptions and identify where evidence diverges.
  - b. User Story 2 - As a Certus Administrator, I want access to detailed provenance and integrity evidence, including source Merkle roots, transformation attestations and continuity checks, so that I can audit the evidence supporting each integrity classification.

### 2.6 User Journeys

The user journeys are:

- i. The user journey for Contact Centre Administrator is:

| Stage                       | User Action                                    | System Response                                                                                                                                     |
|-----------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trust Overview</b>       | Reviews call integrity metrics                 | Displays calls verified and integrity classification distribution                                                                                   |
| <b>Call Records</b>         | Searches or filters calls                      | Displays Call ID, Provenance ID, transformation details and integrity classification                                                                |
| <b>Verification Detail</b>  | Opens a call record                            | Displays relevant provenance, transformation and integrity evidence                                                                                 |
| <b>Integrity Assessment</b> | Reviews the evidence behind the classification | Explains whether the outcome is Authorised Transformation, Benign Channel Variation, Integrity Indeterminate or Suspected Unauthorised Manipulation |
| <b>Investigation</b>        | Reviews an integrity exception                 | Displays available evidence indicating where integrity diverged                                                                                     |

- ii. The user journey for Certus Administrator is:

| Stage                      | User Action                                    | System Response                                                                                      |
|----------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Platform Overview</b>   | Reviews verification activity across customers | Displays aggregate verification and integrity metrics                                                |
| <b>Verification Log</b>    | Searches or filters records across customers   | Displays contact centre, call, transformation and integrity information                              |
| <b>Verification Detail</b> | Opens a verification record                    | Displays detailed provenance and integrity evidence                                                  |
| <b>Evidence Review</b>     | Examines source and transformation evidence    | Displays source Merkle root, transformation attestations, continuity checks and final-media evidence |

|               |                                |                                                                                |
|---------------|--------------------------------|--------------------------------------------------------------------------------|
| Investigation | Reviews an integrity exception | Displays available RCA evidence and the basis for the integrity classification |
|---------------|--------------------------------|--------------------------------------------------------------------------------|

2.7 User Flows

The user flows are:

**i. Contact Centre Administrator**

Trust Overview → Review Integrity Classification Distribution → Open Call Records → Search/Filter Calls → Open Verification Detail → Review Provenance, Transformation and Integrity Evidence → Review Integrity Assessment

**ii. Certus Administrator**

Platform Overview → Review Cross-Customer Verification Metrics → Open Verification Log → Search/Filter Records → Open Verification Detail → Review Detailed Provenance and Integrity Evidence → Review Classification Basis and RCA Evidence

2.8 Accessibility and Responsive Design

Certus uses clear visual hierarchy, readable typography, distinct status labels and adequate spacing to support evidence-dense verification workflows. Integrity outcomes are communicated through text labels alongside colour, reducing reliance on colour alone. Role-based views limit unnecessary technical detail, while dense verification records use structured layouts and horizontally scrollable tables. The MVP is desktop-first, with broader responsive and accessibility validation planned for production.

3. Technical Architecture

Technical Architecture is described in this section.

3.1 Cryptography Glossary and Justifications

[Redacted. Available upon request.]

### 3.2 Technical Architecture Overview

[Redacted. Available upon request.]

### 3.3 Architectural Design Principles

[Redacted. Available upon request.]

### 3.4 End to End Workflow

[Redacted. Available upon request.]

### 3.5 Source Media and Telephony

[Redacted. Available upon request.]

### 3.6 Provenance Generation and Cryptographic Fingerprinting

[Redacted. Available upon request.]

#### 3.6.1 Chunk Construction

[Redacted. Available upon request.]

#### 3.6.2 Final Partial Chunk Handling

[Redacted. Available upon request.]

#### 3.6.3 Provenance Binding

[Redacted. Available upon request.]

#### 3.6.4 SHA-256 Chunk Fingerprinting

[Redacted. Available upon request.]

#### 3.6.5 Ordered Chunk Fingerprints

[Redacted. Available upon request.]

#### 3.6.6 Provenance Stage Output

## **CERTUS PRD**

[Redacted. Available upon request.]

### **3.7 Source Merkle Root**

[Redacted. Available upon request.]

#### **3.7.1 Merkle Tree Input**

[Redacted. Available upon request.]

#### **3.7.2 Merkle Tree Construction**

[Redacted. Available upon request.]

#### **3.7.3 Odd Number of Hashes**

[Redacted. Available upon request.]

#### **3.7.4 Why Certus Generates a Source Merkle Root**

[Redacted. Available upon request.]

#### **3.7.5 Why the Source Merkle Root Is Carried Forward**

[Redacted. Available upon request.]

#### **3.7.6 What the Source Merkle Root Does Not Establish**

[Redacted. Available upon request.]

#### **3.7.7 Merkle Stage Output**

[Redacted. Available upon request.]

### **3.8 Transformations and Signed Attestations**

[Redacted. Available upon request.]

#### **3.8.1 Transformation Chain**

[Redacted. Available upon request.]

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### 3.8.2 Transformation Attestation

[Redacted. Available upon request.]

### 3.8.3 Transformation Continuity

[Redacted. Available upon request.]

### 3.8.4 Digital Signing and Signature Verification

[Redacted. Available upon request.]

### 3.8.5 Scope of a Valid Digital Signature

[Redacted. Available upon request.]

### 3.8.6 Final Expected Media State and Expected Final Hash

[Redacted. Available upon request.]

## 3.9 Independent Verification and Classifications

[Redacted. Available upon request.]

### 3.9.1 Independent Verification

[Redacted. Available upon request.]

### 3.9.2 Root-Cause Analysis

[Redacted. Available upon request.]

### 3.9.3 Integrity Classification

[Redacted. Available upon request.]

### 3.9.4 Verification Record

[Redacted. Available upon request.]

## 3.10 Orchestration, APIs and Persistence

[Redacted. Available upon request.]

## **CERTUS PRD**

### **3.10.1 n8n Orchestration**

[Redacted. Available upon request.]

### **3.10.2 Backend API Layer**

[Redacted. Available upon request.]

### **3.10.3 Workflow Data Continuity**

[Redacted. Available upon request.]

### **3.10.4 Persistence**

[Redacted. Available upon request.]

## **3.11 Front End Integration**

Certus demo portal can be accessed here: [Certus MVP Demo Portal](#)

[Redacted. Available upon request.]

## **3.12 Testing Plan**

[Redacted. Available upon request.]

## **3.13 Acceptance Criteria**

[Redacted. Available upon request.]

## **3.14 Known Limitations**

[Redacted. Available upon request.]

## **3.15 Enterprise Readiness: Reliability, Security and Compliance**

[Redacted. Available upon request.]

# **4. Product Strategy**

## **4.1 Segmentation, Targeting and Positioning (STP)**

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The STP for Certus is as follows:

| STP          | Certus Product Hypothesis                                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Segmentation | Enterprise contact-centre operators, CCaaS platforms and AI speech-processing providers operating workflows in which speech is intentionally transformed.               |
| Targeting    | Initial validation will focus on contact-centre and speech-processing stakeholders involved in deploying, managing or evaluating AI-processed speech and its integrity. |
| Positioning  | Certus is an observational provenance layer that continuously records and verifies speech integrity across AI-processed speech pipelines.                               |

### 4.2 Lean Model Canvas

Based on the current product hypothesis, the Lean Model Canvas in tabular form is as follows:

| Canvas Block             | Certus Application                                                                                                                                                                                                                                   |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Problem                  | Continuous, verifiable provenance is an emerging challenge for AI-transformed contact-centre speech, as legitimate transformations and ordinary telephony effects can both alter the media, thus obscuring its transformation history and integrity. |
| Customer Segments        | Enterprise contact-centre operators, CCaaS platforms and AI speech-processing providers operating workflows in which speech is intentionally transformed.                                                                                            |
| Early Adopters           | To be validated through customer discovery and design-partner pilots.                                                                                                                                                                                |
| Unique Value Proposition | Certus provides continuous media provenance and integrity verification across the lifecycle of AI-processed speech.                                                                                                                                  |
| Solution                 | Source provenance, signed transformation attestations, continuous integrity verification, root-cause analysis and role-based evidence presentation.                                                                                                  |
| Channels                 | To be validated through customer discovery and design-partner pilots.                                                                                                                                                                                |
| Revenue Streams          | To be validated after customer need, business value and willingness to pay are established.                                                                                                                                                          |
| Cost Structure           | Engineering, customer research and design-partner pilots, cloud and media-processing infrastructure, cryptographic infrastructure, security and compliance, and enterprise integrations.                                                             |
| Key Metrics              | Integrity Determination Accuracy (North Star), End-to-End Provenance Continuity, Misclassification Rate, Latency, Design-Partner Pilot Conversion Rate                                                                                               |
| Unfair Advantage         | A provider-agnostic provenance framework designed to maintain continuous evidence across multiple speech transformations and processing stages.                                                                                                      |

The Lean Model Canvas in image form is as follows:

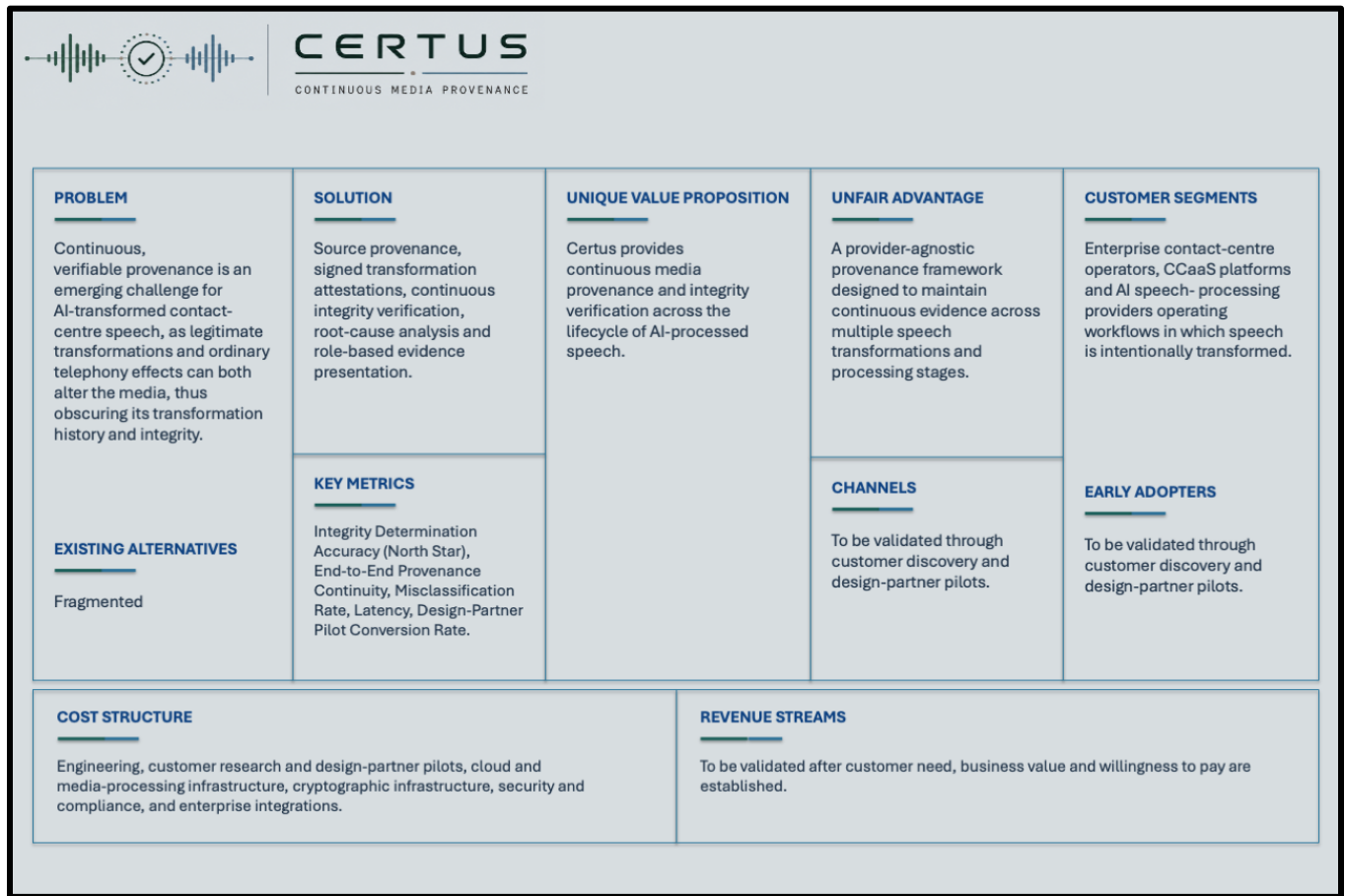


Fig 4.2.1: Lean Model Canvas for Certus

### 4.3 Product Metrics

The currently identified product metrics are:

| Metric Type          | Metric                           | Why it Matters                                                                                                                    |
|----------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Product - North Star | Integrity Determination Accuracy | Measures the proportion of integrity determinations that Certus classifies correctly                                              |
| Product              | End-to-End Provenance Continuity | Measures whether continuous provenance is maintained across the entire speech-processing lifecycle.                               |
| Product              | Misclassification Rate           | Measures the proportion of integrity determinations that Certus classifies incorrectly.                                           |
| Product              | Latency                          | Measures processing latency across the real-time media workflow, with a target of <150 ms one way to support real-time operation. |

|                          |                                             |                                                                                                                                   |
|--------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Business - Future</b> | <b>Design-Partner Pilot Conversion Rate</b> | Measures the proportion of completed design-partner pilots that convert into paying customers or commit to commercial deployment. |
|--------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|

#### 4.4 Feature Prioritisation

The following features were chosen for implementation in the MVP. A combination of MoSCoW and Value vs Effort was chosen for prioritisation.

| Feature                                   | Description                                                                                                                    | MoSCoW           | Value vs Effort                   | Reason                                                               |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------|----------------------------------------------------------------------|
| <b>Source Media Validation</b>            | Validate source media before provenance generation.                                                                            | <b>Must Have</b> | <b>High Value / Low Effort</b>    | Establishes a valid input for provenance processing.                 |
| <b>Provenance Chunking &amp; SHA-256</b>  | Generate 2-sec provenance chunks and cryptographic fingerprints.                                                               | <b>Must Have</b> | <b>High Value / Medium Effort</b> | Establishes granular source provenance.                              |
| <b>Merkle Commitment</b>                  | Generate a Source Merkle Root from ordered chunk fingerprints.                                                                 | <b>Must Have</b> | <b>High Value / Low Effort</b>    | Creates a compact commitment to source provenance.                   |
| <b>Transformation Chain</b>               | Execute the three speech transformations in sequence.                                                                          | <b>Must Have</b> | <b>High Value / High Effort</b>   | Demonstrates provenance across successive media changes.             |
| <b>Signed Transformation Attestations</b> | Record each transformation through signed evidence.                                                                            | <b>Must Have</b> | <b>High Value / Medium Effort</b> | Preserves attributable evidence of transformations.                  |
| <b>Independent Verification</b>           | Verify signatures, transformation order, continuity and provenance context.                                                    | <b>Must Have</b> | <b>High Value / Medium Effort</b> | Verifies the completed provenance and transformation-evidence chain. |
| <b>RCA &amp; Four-Way Classification</b>  | Determine Authorised Transformation, Benign Channel Variation, Integrity Indeterminate or Suspected Unauthorised Manipulation. | <b>Must Have</b> | <b>High Value / Medium Effort</b> | Provides evidence-based integrity outcomes and explains mismatches.  |
| <b>Role-Based Dashboards</b>              | Present evidence through Contact Centre                                                                                        | <b>Must Have</b> | <b>High Value /</b>               | Makes technical evidence                                             |

|                                     |                                                                                             |                           |                                   |                                                                                                     |
|-------------------------------------|---------------------------------------------------------------------------------------------|---------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------|
|                                     | Administrator and Certus Administrator views.                                               |                           | <b>Medium Effort</b>              | understandable at the intended role level.                                                          |
| <b>Frontend–Backend Integration</b> | Connect the role-based frontend to the executable Certus backend and verification workflow. | <b>Won't Have for now</b> | <b>High Value / Medium Effort</b> | Enables verification evidence and results to be surfaced dynamically through the product interface. |
| <b>Live Telephony Integration</b>   | Integrate Certus with live streaming media.                                                 | <b>Won't Have for now</b> | <b>High Value / High Effort</b>   | Required to validate Certus beyond controlled file-based execution in a live environment.           |

#### 4.5 Build, Buy, Integrate Strategy

The strategy is as follows:

| <b>Decision</b>         | <b>Capabilities</b>                                                                                                                    | <b>Rationale</b>                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Build Internally</b> | Provenance model, media identity and chunk binding, verification logic, integrity classification, attestation framework and RCA logic. | These capabilities form the core Certus product and its provenance-verification approach.              |
| <b>Buy / Subscribe</b>  | Cloud infrastructure, managed database and storage, identity and access services, observability, and KMS/HSM services.                 | Established services reduce the need to build commodity production infrastructure internally.          |
| <b>Integrate</b>        | CCaaS platforms and telephony/media interfaces.                                                                                        | Enables Certus to operate within production communication workflows while remaining provider-agnostic. |

#### 4.6 Customer Service Strategy

A detailed customer service and support model has not been defined at the current MVP stage. Production deployment patterns, operational support needs, incident types and service-level requirements require validation through primary research.

Customer discovery and design-partner pilots will inform how Certus should be supported in real-world deployments. Support tiers, response commitments and service-level targets will be defined based on this evidence.

## 5. Marketing Plan

A marketing and go-to-market plan has not been defined at the current MVP stage. Certus has established the technical feasibility of its continuous provenance and integrity-verification approach; however, customer demand, business value, target buyer, willingness to pay and product-market fit have not yet been validated through primary research.

The next stage will focus on customer discovery followed by design-partner pilots with relevant contact-centre, CCaaS and speech-processing stakeholders. Customer discovery will validate the problem, target buyer and perceived business value, while design-partner pilots will evaluate whether Certus' provenance and integrity evidence provides meaningful operational value in real-world workflows. Marketing, pricing, channel and go-to-market decisions will be defined only after sufficient evidence is available to support them.

## 6. Ethical Considerations

Certus is designed to avoid declaring transformed speech trustworthy or malicious without sufficient evidence. Ethical operation requires privacy, data minimisation, evidence-based classification, controlled access to sensitive media and explicit communication of uncertainty. Integrity Indeterminate is therefore a deliberate outcome when available evidence cannot support a reliable classification.

| Ethical Consideration | Certus Approach                                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Privacy          | Process call media only when authorised; minimise retained media and sensitive content.                                                                      |
| Transparency          | Surface the evidence supporting each integrity classification.                                                                                               |
| Uncertainty           | Use <b>Integrity Indeterminate</b> when evidence is insufficient for a reliable classification.                                                              |
| Human Oversight       | Do not use Certus classifications as the sole basis for punitive employee or customer decisions.                                                             |
| Attribution           | Distinguish authorised transformations, benign channel variation and suspected unauthorised manipulation based on available evidence rather than assumption. |
| Evidence Protection   | Restrict access to provenance records, fingerprints, Merkle commitments, signed attestations and verification evidence according to role and need-to-know.   |

## 7. Future Roadmap

Certus establishes continuous provenance and integrity verification across the lifecycle of AI-processed speech. As Certus matures, the future roadmap extends this foundation through intelligent investigation capabilities and, subsequently, provenance verification across multiple media modalities.

| Version                         | Capability                                | Primary Function                                                                                                                                                                                                          |
|---------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>V1 —<br/>Current<br/>MVP</b> | Continuous<br>Speech<br>Provenance        | Establish source provenance, preserve transformation evidence across AI-based speech processing, verify media integrity and classify resulting integrity outcomes.                                                        |
| <b>V2</b>                       | AI-Assisted<br>Integrity<br>Investigation | Extend integrity verification with change localisation to specific timestamps or speech segments, and use AI to summarise provenance evidence, correlate integrity events and assist investigators with anomaly analysis. |
| <b>V3</b>                       | Multimodal<br>Provenance                  | Extend the Certus provenance model beyond speech to other AI-processed enterprise media such as video and multimodal communication.                                                                                       |

## 8. Project Management

The project followed an iterative build-and-validate approach. Work began with architecture and audio-processing specifications, followed by provenance design and modular Python implementation. Transformations, cryptographic attestations, integrity verification, RCA and classification were then developed and validated alongside n8n orchestration, followed by role-based frontend demonstration. This approach enabled progressive validation of the end-to-end provenance and integrity model.

The project plan implemented for the MVP is as follows:

| Sprint                                                                             | Duration  | Objective                                                                 | Key Activities                                                                                                                                                                   | Deliverables                                                                             |
|------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Sprint 1 –<br/>Architecture<br/>&amp;<br/>Provenance<br/>Foundation</b>         | Week 1    | Establish audio-processing specifications and source provenance           | Set up Python/Flask and n8n; validate 16 kHz audio; implement 2-sec provenance chunks, SHA-256 fingerprints and Merkle commitment                                                | Working provenance foundation and stable identifiers                                     |
| <b>Sprint 2 –<br/>Transformations,<br/>Attestations<br/>&amp;<br/>Verification</b> | Weeks 2–3 | Build and orchestrate the transformation and integrity-verification chain | Integrate DeepFilterNet3, SpeechBrain SepFormer and Seed-VC V2; implement Ed25519 signed attestations; develop n8n orchestration, integrity verification, RCA and classification | End-to-end provenance, transformation evidence, verification and four-way classification |
| <b>Sprint 3 –<br/>Frontend &amp;<br/>Product<br/>Validation</b>                    | Week 4    | Translate technical evidence into role-based product experience           | Build Contact Centre Administrator and Certus Administrator views; surface provider, transformation, verification and RCA evidence; validate the MVP and finish documentation    | Demonstrable Certus MVP and product documentation                                        |

-----End of the Document-----